

2018~2019 学年第 2 学期期末考试试卷
《形式语言与自动机》(A 卷 共 5 页)
(考试时间: 2019 年 6 月 15 日)

题号	一	二	三	四	五	成绩	核分人签字
得分							

一、Formal Languages (20 分)

1. (5 分) Given $\Sigma_1 = \{a, b\}$, $\Sigma_2 = \{0, 1\}$, $L_1 = \{ab, ba, bb\}$, $L_2 = \{00, 11\}$, then
 $\Sigma_1^* = \{ \rule{10cm}{0.4pt} \}$
 $\Sigma_2^+ = \{ \rule{10cm}{0.4pt} \}$
 $L_1 L_2 = \{ \rule{10cm}{0.4pt} \}$

2. (5 分) Given a language $L = \{ww^R \mid w \in \{a,b\}^+\}$, please construct a grammar
 $G = (V, T, P, S)$ such that $L(G) = L$

$V = \{ \rule{2cm}{0.4pt} \}$, $T = \{ \rule{2cm}{0.4pt} \}$, $S = \{ \rule{1cm}{0.4pt} \}$
 $P = \{ \rule{3cm}{0.4pt} \}$
 $\rule{3cm}{0.4pt}$
 $\rule{3cm}{0.4pt}$
 $\rule{3cm}{0.4pt} \}$

3. (10 分) Given the grammar $G = (\{S, B\}, \{a, b, c\}, P, \{S\})$ where P is defined as
 $S \rightarrow aBc$
 $S \rightarrow aSBc$
 $cB \rightarrow Bc$
 $aB \rightarrow ab$
 $bB \rightarrow bb$
Please prove that $L(G) = \{a^n b^n c^n \mid n \geq 1\}$

二、Finite Automata (45 分)

4. (5 分) The formal description of a DFA M is $(\{q_1, q_2, q_3, q_4, q_5\}, \{u, d\}, \delta, q_3, \{q_3\})$, where δ is given by the following table. Give the state diagram of this machine.

	u	d
q_1	q_1	q_2
q_2	q_1	q_3
q_3	q_2	q_4
q_4	q_3	q_5
q_5	q_4	q_5

5. (5 分) Give the state diagram of a DFA recognizing the following language where $\Sigma = \{a, b\}$.

$\{w \mid w \text{ has exactly two } a\text{'s and at least two } b\text{'s}\}$

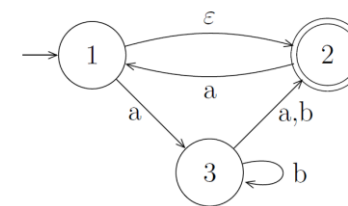
6. (5 分) Give the state diagram of a DFA recognizing the following language where $\Sigma = \{a, b\}$.

$\{w \mid w \text{ does not contain the substring } baba \}$

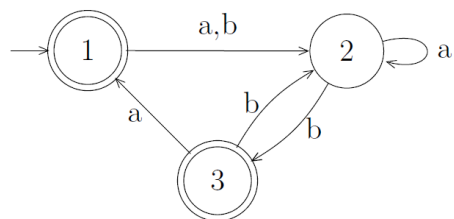
7. (5 分) Give the state diagram of an NFA with three states recognizing the following language where $\Sigma = \{a, b\}$.

$\{w \mid w \text{ ends with } 00 \}$

8. (5 分) Convert the following NFA to an equivalent DFA.



9. (10 分) Convert the following DFA to a regular expression.



10. (10 分) Convert the following regular expression to an NFA.

$(a \cup b)^* aba$

三、Pumping Lemma (10 分)

11. (10 分) Use the pumping lemma to prove that the following language is not regular.

$\{w \mid w \in \{0, 1\}^* \text{ is not a palindrome } \}$

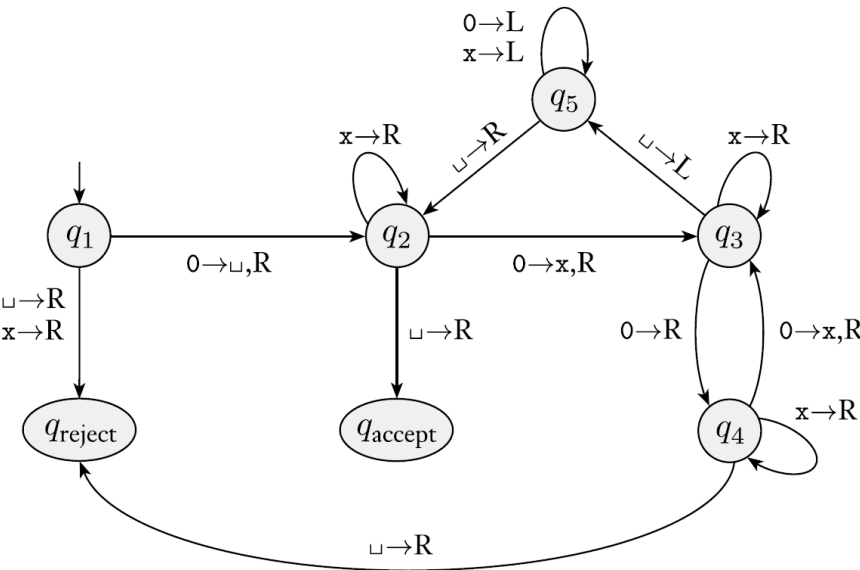
(Hint: 1. A palindrome is a string that reads the same forward and backward.

2. Use the closure of the class of regular languages under complement.)

四、Turing Machines (15 分)

12. (5 分) A Turing Machine is a 7-tuple $(Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ where Q is the set of states,
 Σ is _____ not containing blank symbol $\sqcup$,
 Γ is _____ where _____,
 δ : _____ is the transition function,
 $q_0 \in Q$ is the start state,
 $q_{\text{accept}} \in Q$ is the accept state, and
 $q_{\text{reject}} \in Q$ is reject state, where _____.
13. (10 分) Given the state diagram of the Turing machine that recognizes the language

$\{0^{2^n} \mid n \geq 0\}$



Give a run of this machine on input string 0000 by writing out a sequence of configurations. The first configuration in the sequence is the start configuration q_10000 , and the last configuration in the sequence is the accepting configuration $\sqcup xxx \sqcup q_{\text{accept}}$. Please write out the remaining configurations in the sequence.

- | | | | |
|-----------------------------|-----------------------------|-----------------------------|---|
| ① q_10000 | ⑦ _____ | ⑬ $\sqcup xxx q_3 \sqcup$ | ⑲ _____ |
| ② $\sqcup q_2000$ | ⑧ _____ | ⑭ $\sqcup xx q_5 x \sqcup$ | ⑳ _____ |
| ③ _____ | ⑨ $q_5 \sqcup x 0 x \sqcup$ | ⑮ $\sqcup x q_5 x x \sqcup$ | ㉑ _____ |
| ④ $\sqcup x 0 q_4 0$ | ⑩ $\sqcup q_2 x 0 x \sqcup$ | ⑯ _____ | ㉒ $\sqcup xxx \sqcup q_{\text{accept}}$ |
| ⑤ _____ | ⑪ _____ | ⑰ $q_5 \sqcup xxx \sqcup$ | |
| ⑥ $\sqcup x 0 q_5 x \sqcup$ | ⑫ $\sqcup xx q_3 x \sqcup$ | ⑱ _____ | |

五、Proofs (10 分)

14. (10 分) An **all-NFA** M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ that accepts $x \in \Sigma^*$ if *every* possible state that M could be in after reading input x is a state from F . Note, in contrast, that an ordinary NFA accepts a string if *some* state among these possible states is an accept state. Prove that all-NFAs recognize the class of regular languages.